

Appendix: The forward / inverse maps $(\nu, \delta) \leftrightarrow (N, \Delta)$ — canonicalization by the conserved quantity, a unified map, the privileged scale, and full invertibility

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Purpose

This appendix gives a **rigorous formulation, in a single unified form, of the forward and inverse maps** between the observable side $(\nu, \delta)_n$ and the fundamental-quantity side $(N, \Delta)_n$. The key is to **canonicalize (normalize) by the conserved quantity** on the right-hand side of the identity. As a result:

1. Every anonymous quantity (the central value ν , the fluctuation δ , the conserved value on the right-hand side and its fluctuation) is mapped by the **same map** $\text{func}(\text{value/normalization})$.
2. The **scale (the conserved quantity) used for normalization is the only privileged (non-anonymous) coordinate that is passed unchanged**.
3. The inverse map also has the same form, $\text{func}^{-1}(\text{mapped value}) \times (\text{normalization})$, so **both the individual values and the fluctuations are recovered exactly**.

This is a discrete system; the continuum is not treated. $1/2$ is a posit (parent paper §6.2). Interpretive matters such as the conjugacy of the roles of value and uncertainty are not treated (out of scope).

1. Domain (discrete; exact range)

Following §4 of the parent paper.

- Number of axes $n \in \mathbb{Z}_{\geq 1}$.
- Fundamental quantity $N_i \in \mathbb{Z}$ (integers; parent paper §4: “when $\nu = 2^n$, $N = n$, and N ranges over all integers $\mathbb{Z}$ ”).

- Central value $\nu_i = 2^{N_i}$ (**powers of 2**, $\nu_i \in \{2^m : m \in \mathbb{Z}\}$).
- The fluctuation δ_i is expressed as a multiplicative factor $\varphi_i = 2^{\Delta_i} > 0$ ($\Delta_i \in \{\pm\frac{1}{2}\}$, the half-bit, a posit). That is, the value including the fluctuation is $\nu_i \varphi_i = 2^{N_i + \Delta_i}$.

Identity (conserved quantity): as in §5 of the parent paper,

$$\prod_i \nu_i = k \iff \sum_i N_i = K \ (= \log_2 k).$$

K (or k) is the **conserved quantity on the right-hand side**.

2. The principle of the unified map

Take the core of the map to be

$$\text{func} = \log_2, \quad \text{func}^{-1} = 2^{(\cdot)}.$$

Define the **normalization (the privileged scale)** as the per-axis value of the conserved quantity,

$$C = k^{1/n} = 2^{K/n}$$

(so $\log_2 C = K/n$).

- **Forward map (same form for every anonymous quantity):** normalize the quantity x by a reference r ,

$$X = \text{func}\left(\frac{x}{r}\right) = \log_2 \frac{x}{r}.$$

- **Inverse map (same form throughout):**

$$x = r \cdot \text{func}^{-1}(X) = r \cdot 2^X = \text{func}^{-1}(X + \log_2 r).$$

(The right-most form folds the normalization inside func^{-1} .)

- **The privileged scale C is the only coordinate passed unchanged** (it is the basis of normalization and is not anonymous). Every other quantity is mapped by the same func as a ratio against C (or a lower-level reference).

3. The map of the central value ($\nu \leftrightarrow N$, normalized form)

With reference $r = C$, normalize and map the central value:

$$\hat{N}_i = \text{func}\left(\frac{\nu_i}{C}\right) = \log_2 \frac{\nu_i}{C} = N_i - \frac{K}{n}.$$

The effect of normalization:

$$\sum_i \hat{N}_i = \sum_i N_i - n \cdot \frac{K}{n} = K - K = 0, \quad \prod_i \frac{\nu_i}{C} = \frac{k}{C^n} = \frac{k}{k} = 1.$$

That is, **canonicalization makes the conserved quantity on the right-hand side equal to 1 (i.e., 0 in the logarithm)**. $\hat{N}_i$ is the **shape (direction)** of the central value, carrying the individuality of each axis.

Inverse map (same form):

$$\nu_i = C \cdot \text{func}^{-1}(\hat{N}_i) = C \cdot 2^{\hat{N}_i} = 2^{\hat{N}_i + K/n} = 2^{N_i}.$$

The pair $(C, \hat{N})$ is in **bijection** with (N_i) (since $N_i = \hat{N}_i + K/n$).

4. The map of the fluctuation ($\delta \leftrightarrow \Delta$, same form)

Viewing the fluctuation as a multiplicative factor $\varphi_i = 2^{\Delta_i}$ (reference $r = 1$), map it by the same func:

$$\Delta_i = \text{func}(\varphi_i) = \log_2 \varphi_i \in \{\pm \frac{1}{2}\}, \quad \varphi_i = \text{func}^{-1}(\Delta_i) = 2^{\Delta_i}.$$

That is, **the fluctuation obeys exactly the same map rule as the central value** (by anonymity, ν and δ share the same forward map, and N and Δ share the same inverse map). On a conjugate pair we impose sum-zero $\sum_i \Delta_i = 0$ (parent paper §6.3).

The full quantity including the fluctuation is $\nu_i \varphi_i = C \cdot 2^{\hat{N}_i + \Delta_i}$; conversely $\hat{N}_i + \Delta_i = \log_2(\nu_i \varphi_i / C)$.

5. The right-hand side (the conserved quantity) and the privileged scale

The quantity on the right-hand side of the identity fits in the same frame:

- **The conserved value $k \rightarrow 1$ after normalization (i.e., 0 in the logarithm).** The value itself is **passed unchanged** as the privileged scale $C = k^{1/n}$.
- **The fluctuation of the conserved value $\rightarrow 0$.** By the sum-zero of the conjugate pair $\sum \Delta_i = 0$, the product including fluctuations is still $\prod(\nu_i \varphi_i) = k \cdot 2^{\sum \Delta_i} = k$, invariant. Hence the right-hand fluctuation maps $0 \rightarrow 0$.

Therefore the right-hand side (the conserved quantity) is **the basis of normalization = the privileged scale**: (i) its value is canonicalized to 1, (ii) its fluctuation is 0, (iii) only the scale C is passed unchanged. This is what “the one extra (+1) dimension that is passed unchanged is the only non-anonymous coordinate” really means.

6. Full invertibility (both individual values and fluctuations are recovered exactly)

The data retained are $(C, \hat{N}, \Delta) = (\text{privileged scale, central shape, fluctuation})$. By the inverse map (the same form as §2–§4),

$$\nu_i = C \cdot 2^{\hat{N}_i} = 2^{N_i}, \quad \varphi_i = 2^{\Delta_i},$$

so **the individual central values ν_i and the fluctuations Δ_i (δ_i) are recovered exactly.** $(N_i) \leftrightarrow (C, \hat{N}_i)$ is a bijection, and so is $\Delta_i \leftrightarrow \varphi_i$. **This canonicalized map is fully invertible.**

Even if a non-integer (e.g., $C = 2\sqrt{2}$) appears, it is **absorbed by the privileged scale C** , while the structure (the shape $\hat{N}$, the fluctuation Δ) stays discrete and clean, remaining within the domain.

7. The canonicalization that discards the scale (gauge-fixing of g)

The scale C is a privileged coordinate. If it is retained, the map is fully invertible as in §6. If instead it is **discarded and one works with $(\hat{N}, \Delta)$ only, what remains is a scale-invariant shape + fluctuation = the structure (the ω side)**. This matches the g/ω separation of §7.3 of the parent paper (the scale g can be removed as a gauge, while the quantization ω remains invariant). In either case the map preserves the structure $(\hat{N}, \Delta)$ (retain C and the individual values are fully recovered; discard C and the scale-invariant structure remains).

8. Numerical check (an actual sequence of operations)

Input $n = 2$, $\nu = (2, 4)$, fluctuation $\Delta = (+\frac{1}{2}, -\frac{1}{2})$ (sum-zero).

1. **Conserved quantity / scale:** $k = \prod \nu_i = 8$, $K = \sum N_i = 1 + 2 = 3$, $C = k^{1/2} = 2\sqrt{2} \approx 2.828$ (the privileged scale).
2. **Forward map (central value, normalized):** $\hat{N}_i = \log_2(\nu_i/C)$: $\hat{N}_1 = \log_2(2/2\sqrt{2}) = \log_2(2^{-1/2}) = -\frac{1}{2}$, $\hat{N}_2 = \log_2(4/2\sqrt{2}) = \log_2(2^{+1/2}) = +\frac{1}{2}$. $\rightarrow \hat{N} = (-\frac{1}{2}, +\frac{1}{2})$, $\sum \hat{N} = 0$.
3. **Forward map (fluctuation):** $\varphi_i = 2^{\Delta_i} = (2^{+1/2}, 2^{-1/2}) = (\sqrt{2}, 1/\sqrt{2})$.
4. **Inverse map (central value):** $\nu'_i = C \cdot 2^{\hat{N}_i} = 2\sqrt{2} \cdot (2^{-1/2}, 2^{+1/2}) = (2, 4) \leftarrow$ **recovered exactly.**
5. **Inverse map (fluctuation):** $\Delta'_i = \log_2 \varphi_i = (+\frac{1}{2}, -\frac{1}{2}) = \Delta \leftarrow$ **invariant.**

Result: $\nu' = (2, 4) = \nu$, $\Delta' = \Delta$. **Both the individual values and the fluctuations are**

conserved exactly. The non-integer $2\sqrt{2}$ is contained in the privileged scale C , while the shape $(-\frac{1}{2}, +\frac{1}{2})$ and the fluctuation $(\pm\frac{1}{2})$ stay clean.

9. Summary

Quantity	Reference (normalization)	Forward map $\text{func}(x/r)$	Inverse map $r \cdot \text{func}^{-1}(X)$
Central value ν_i	Scale $C = k^{1/n}$	$\hat{N}_i = \log_2(\nu_i/C)$ ($\sum \hat{N} = 0$)	$\nu_i = C \cdot 2^{\hat{N}_i}$
Fluctuation φ_i	1	$\Delta_i = \log_2 \varphi_i$ ($\sum \Delta = 0$)	$\varphi_i = 2^{\Delta_i}$
Right-hand conserved value k	k	$1 \rightarrow \log_2 1 = 0$	passed unchanged as scale C (privileged)
Right-hand fluctuation	—	0 (invariant by sum-zero)	0

- **Same map:** every anonymous quantity goes through $\text{func}(x/r) = \log_2(x/r)$, with inverse $r \cdot 2^X$.
- **Only the privileged scale C** is passed unchanged — the single non-anonymous coordinate (the +1 dimension).
- **Fully invertible:** the individual ν_i and Δ_i are recovered exactly from $(C, \hat{N}, \Delta)$; non-integers are absorbed by the scale.
- **Gauge-fixing** = the canonicalization that discards the scale C (retaining only the scale-invariant structure $(\hat{N}, \Delta)$; the g/ω separation, §7.3).

10. Limitations (notes for reading)

1. $1/2$ is a **posit** (parent paper §6.2).
2. **A discrete system:** $N \in \mathbb{Z}$, $\nu = 2^N$. All non-integers are contained in the privileged scale C , and the structure $(\hat{N}, \Delta)$ is discrete.
3. **The fluctuation is treated as a multiplicative factor** $\varphi = 2^\Delta$ (so that it rides on the same func as the central value). The signed addition $\pm\frac{1}{2}$ corresponds to $\Delta = \log_2 \varphi$.
4. **The scale C is a privileged (non-anonymous) coordinate.** The canonicalization that discards it corresponds to scale-invariance (gauge-fixing of g) and is consistent with §7.3.
5. **No physical identification is made.**

References

- Kihara, N., “On the Logarithmic Map of Scale and the Separation of Scale and Quantization in the High-Symmetry Limit — A Limited Verification Report from a Minimal Axiom System (Revised Edition),” §4, §5, §6.3, §7, Axioms 2, 4, 5. Concept DOI: 10.5281/zenodo.20740841.